Linguistic transparency as mediated accessibility

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Abstract

This paper analyzes linguistic transparency as a family of mediated accessibility relations across syntax and form–meaning mapping. A structure is transparent when a property remains accessible through a grammatical or representational mediator for some relation, despite structure that might otherwise conceal it. The resulting schema is a comparative concept: it asks which local profiles support reliable projection from observable configurations to less visible grammatical properties. Kind-like behaviour belongs to individual profiles.

The worked cases are English number-transparent quantificational nouns and transparent free relatives, with comparative extensions to Spanish, Tsez, Aleut, Lardil, Kayardild, Chamorro, and Algonquian/Kutenai obviation. Form–meaning accessibility, including constructional compositionality, compound transparency, and morphological priming, shows how the same schema reaches beyond syntax. The result is a field-relative model in which grammatical kindhood is a downstream verdict of projectibility, not an inherent structural feature.

Introduction

The initial puzzle is that some grammatical relations can still access a property through a structure that appears, on the surface, to block or reverse that access. Two examples from Huddleston and Pullum's *Cambridge Grammar of the English Language* (*CGEL*, 2002) make the point visible (§18.2).

(1) a. *A number of spots have/*has appeared.*
 b. *Heaps of money has/*have been spent.*

In the first example, the surface head *number* is singular, but agreement is plural because the *of*-complement *spots* is plural count. In the second, the surface head *heaps* is plural, but agreement is singular because the *of*-complement *money* is non-count. The construction lets the complement's number/count profile reach the verb, even though the surface head's own morphological number points the other way.

CGEL calls these NUMBER-TRANSPARENT QUANTIFICATIONAL NOUNS (QNs). Core members include *lot, plenty, lots, bags, heaps, loads, oodles, stacks, remainder, rest, number,* and *couple*, with *majority* and *minority* at the boundary. They head NPs whose agreement override is, on *CGEL*'s analysis,

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the default, unlike the optional override of ordinary collectives. The wider question is whether comparable *transparency* labels pick out real accessibility relations or only restate the facts they describe. The umbrella schema is:

> Umbrella schema 1.
>
> X is *transparent* with respect to property P for relation R when P, which X might otherwise have concealed from R, remains accessible through X for the purposes of R.

Applied to (1a), Schema 1 binds X to the number-transparent quantificational noun + *of*-complement construction, P to the number/count profile of *spots*, and R to subject–verb agreement.

The useful cases are PROJECTIBILITY PROFILES. Projectibility inherits Goodman's question of which predicates support reliable inductive projection (Goodman, 1955); Boyd's maintenance-based realism supplies the stabilizing-process side (Boyd, 1991). A profile has three components: an OBSERVABLE INPUT, predictions the input licenses, and a STABILIZING PROCESS. In the worked example, the input is membership in the number-transparent QN class; the predictions include agreement override, determiner frames, partitive availability, fused-head behaviour, count/non-count percolation, and restricted premodification; the stabilizer is lexical inheritance.

The substantive clause is *remains accessible through X*. A case fits only if X has internal structure whose default behaviour would have made P inaccessible to R. Coordination (§2.3) is the textbook exclusion: ordinary plural coordination has no head whose morphological number competes for agreement with the conjuncts' features, so calling it *transparent* would relabel summation as access. Everyday senses of *transparent prose* or *transparent intentions* are excluded too; the schema's slots have no values to bind.

This non-instance requirement is the umbrella's distinctive contribution. It keeps mediated accessibility from collapsing into a relabeling of familiar facts: the schema earns its place not by naming a kind but by supplying a relational diagnostic that a per-construction analysis lacks, one that separates genuine mediation from compositional summation and from everyday CLARITY. So TRANSPARENCY, as an umbrella, isn't itself projectible: applied on its own, the label projects nothing, and what projects is the particular profile it points to.

The paper therefore asks three questions. Which linguistic *transparency* labels bind a non-trivial X-P-R triple? Which triples support projection beyond the facts that motivated the label? What stabilizes the strongest profiles? The primary field is syntax, where the targets include agreement, syntactic-category licensing, complement selection, and fused-head behaviour. The answer is strongest for number-transparent QNs and transparent free relatives.

The English QN class illustrates the strongest case: once the class is recognized as small and conventional, membership predicts the whole cluster, not just the agreement override, reliably for core members and weakly at the edges (*majority, minority*, sense-split *couple*).

Transparent free relatives fill the slots differently. In *What appear to be trousers are on the chair*, X is the *what*-clause, P is the plural NP nucleus *trousers*, and R is matrix subject–verb agreement. In attributional cases, a second relation appears: the matrix accesses the nucleus's syntactic category through the *what*-clause.

The typological extension then tests the topology of the schema. Spanish and Tsez keep the outward endpoint shape but change the language and, for Tsez, the marking type. Aleut changes the source status: P can be recoverable rather than overt when a missing possessor triggers anaphoric verbal morphology, though Merchant's analysis treats that possessor as syntactically present. Lardil, Kayardild, and Chamorro show that access can also run inward onto dependents or be distributed along a path. Algonquian/Kutenai obviation marks the boundary where a feature is relayed onto a head rather than accessed directly.

Projectibility is field-relative: a profile supports induction for a field's purposes, not for every inquiry at once. X can be a lexical item, construction, intensional context, or derivationally complex word. P can be a number/count feature, constituent meaning, referent, or stem representation. R can be agreement, interpretation, substitution under co-reference, or masked priming. Accessibility itself can be class-like, as in core QNs, or gradient, as in compositionality ratings and priming magnitudes.

My aim is constructive rather than exhaustive. The paper leaves aside most uses of TRANSPARENCY in linguistics, including extraction, islands, barriers, and phases; extraction-transparency is acknowledged in §2 but not developed, and Chamorro wh-agreement appears in §4 only as a topology comparison. It also leaves aside phonological or orthographic transparency and everyday talk of *transparent explanation*. Referential or *de re*/*de dicto* transparency appears in §3.1 as the boundary case where the mediator-form structure stretches thinnest.

MEDIATED ACCESSIBILITY is more specific than CLARITY and broader than COMPOSITIONALITY. Clarity is rhetorical: it doesn't single out a mediator or relation. Compositionality is the limiting case of form–meaning transparency, where a mediator's parts predict the whole. Number transparency is different: a QN's morphological number doesn't by itself predict the verb's number, and under the construction it doesn't have to.

Section 2 takes up morphosyntactic transparency, with number-transparent QNs and transparent free relatives as paradigms. Section 3 turns to form–meaning accessibility (constructional compositionality (Goldberg, 2006), rated compound transparency (Bell & Schäfer, 2016), and time-sensitive lexical processing (Heyer & Kornishova, 2018)), where the relation R shifts from agreement to interpretation and processing, and notes referential transparency as the schema's boundary case; that family is developed in its own right in separate work.

Section 4 applies Haspelmath's (2010) comparative-concept discipline to the schema's cross-linguistic ambitions and introduces the topology map. Section 5 names the stabilizers and treats Slater's stable property cluster (SPC) framework (Slater, 2015) and Khalidi's causal-network account (Khalidi, 2018) as complementary diagnostics.

2 Morphosyntactic transparency

Morphosyntactic transparency is the case where X is a syntactic construction (or a lexical class within one) and R is a matching relation between a non-head property and a co-occurring exponent.[1]

[1] Extraction transparency (whether an island or barrier is opaque or transparent to long-distance dependency formation) is a morphosyntactic-accessibility relation under the schema (with R = dependency formation), and the associated question remains active. Transformational accounts posit empty elements (traces, copies) that mediate the connection between a fronted element and its base position; non-transformational dependency accounts (da Cunha & Gibson, 2025; Gibson, 2025) argue that no such mediating empty elements are needed and that direct head-to-dependent connections better fit acceptability and processing data. Both sides make schema-level claims about whether X contains invisible

Subject-verb agreement is the textbook instance. *CGEL*'s number-transparent quantificational nouns are the paradigm; collective NPs are a transitional case; coordination is a non-instance; transparent free relatives add a second profile, stabilized by constructional rather than lexical inheritance, with two dissociable R-relations under one mediator. Section 2.6 marks the Korean extension as deferred rather than treating it as evidence for the present syntactic result.

The concrete contrasts are simple. In (1a), agreement tracks *spots* through singular *number*; in (9a), agreement tracks *trousers* through singular *what*.

This section carries the paper's primary field-relative claim. The projected targets are syntactic and morphosyntactic: agreement, syntactic-category licensing, complement selection, fused-head behaviour, and distributional restrictions inside NP and fused-relative constructions. Nothing in the syntactic result depends on phrase-structure representation as such: the profiles require only observable relations among a mediator, an internal property, and the grammatical relation that tracks it, whether those relations are represented in constituency or dependency terms.

2.1 Number-transparent quantificational nouns

The paradigm again is *CGEL*'s two-direction contrast in (1): plural agreement despite singular *number*, and singular agreement despite plural *heaps*. Schema-wise, X is the number-transparent quantificational noun + *of*-complement construction; P is the number/count profile of the *of*-complement; R is subject-verb agreement.

CGEL's "simple agreement rule" is head-driven: a singular head NP triggers singular agreement, a plural head NP plural agreement (§18.1). Number-transparent NPs override the rule. *CGEL* describes the override as obligatory in the central cases; rare singular-verb attestations with *number* are noted in footnote 73 to §18.2 and analyzed there as hypercorrections rather than as variant grammar. The contrast with the optional override of collective nouns (§2.2) is the operative point. The construction's job is to let the *of*-complement's number/count profile reach agreement past the head's morphological number, which would otherwise have determined it.

The core inventory is small and well documented. *CGEL* identifies the principal members in example [57]: *lot, plenty* (singular forms; *lot* most often with *a*-determiner, *plenty* typically without a determiner); *lots, bags, heaps, loads, oodles, stacks* (plural forms, typically without a determiner); *remainder, rest* (singular forms typically with *the*-determiner); *number, couple*$_2$ (singular forms typically with *a*-determiner, taking only plural *of*-complements, partitive or pseudo-partitive). Of these, "the clear cases of number-transparent singular nouns are *lot, number*, and *couple*$_2$"; *majority* and *minority* are borderline (*CGEL* [13] notes plural override is "surely obligatory" in some examples but singular agreement is occasionally attested; §18.2). The determiner patterns above are typical, not absolute: *the lot of it* (in a totality sense) is grammatical alongside *a lot of it*, and similar sense-conditioned variations exist for other members.

Keizer's study of English NP categorization and Van Eynde and Kim's Head-driven Phrase Structure Grammar (HPSG) analysis of pseudo-partitives are the immediate background for treating these strings as structured binominal configurations rather than as a single loose family (Keizer, 2007; Van Eynde & Kim, 2023).

Class membership predicts a cluster of behaviours beyond the agreement override. The relevant contrast is the standard PARTITIVE / PSEUDO-PARTITIVE distinction: partitive *of*-complements re-

mediating structure. The centre of the argument here is the agreement-style cases.

quire a definite complement that picks out a subset of a definite set (*of the money*, *of those books*, *of these soldiers*), while pseudo-partitive *of*-complements express a quantity, measure, or collection relation with a bare or otherwise non-definite complement in the QN frame (*of money*, *of soldiers*, *of some paintings*) (Kim & Michaelis, 2020; Selkirk, 1977; Stickney, 2007; Van Eynde & Kim, 2023).[2]

Lot most often takes *a* as determiner with limited premodification (*a whole/huge lot of time*; *CGEL*, §3.3); *the lot of it* is also available in a totality sense, but other determiners are blocked (**some lot of it*). *Plenty* typically takes no determiner or modifier (*plenty of butter*, not generally *a remarkable plenty of butter*), and like the plural-only forms it allows virtually no partitive *of*-complements.[3] The plural-only forms (*lots, bags, heaps, loads, oodles, stacks*) show the same pattern.

(2) a. *They have oodles of money.*
b. *Oodles of the money had already been spent.*

The contrast in (2) shows the plural-only pattern: the first is normal; the second is rare to non-attested. *Remainder* and *rest* take only partitive *of*-complements in their characteristic use; *number* and *couple*$_2$ take plural obliques.

(3) a. *the remainder of the time*
b. **the remainder of time*
c. *a number of the protesters*
d. **a number of money*

The oblique contrasts in (3) are highly structured even where individual cases admit sense-conditioned variation.

The pattern shows what number transparency *is* in the umbrella schema's vocabulary. X (the construction) leaves P (the number/count profile of the *of*-complement) accessible to R (agreement). What X conceals from agreement is the morphological number of the head noun *lot, plenty, heaps*, etc. The agreement-targeting feature is on the *of*-complement, not on the surface head. That's what makes X transparent with respect to P: X lets P reach the verb anyway.

The override is only visible where the head's number and the complement's diverge. *A lot of people are* shows it (singular *lot*, plural *people*); *a lot of money is* doesn't, since head and complement both call for the singular. The discriminating cases are the crossed ones: a singular-form quantificational noun with a plural complement, or a plural-form one (*lots, heaps*) with a non-count complement. The matched cases are silent, not counterevidence. Transparency is therefore under-detected: a class can be transparent while most of its tokens look head-driven, so corpus frequency understates it.

[2] The headedness issue is framework-internal. HPSG analyses treat the first noun as syntactic head in *A of the B* partitives, while *CGEL* retains the quantificational noun as head and treats agreement with the complement as override. Since §3 keeps the umbrella analysis framework-neutral, this paper doesn't adjudicate the headedness choice.

[3] A pilot COCA check (May 2026) across matched complements quantifies the contrast: plural-only forms show $\approx 0.2\%$ partitive uses; *plenty* $\approx 0.03\%$; *rest* / *remainder* show the reverse pattern at $\geq 97\%$ partitive (raw, higher after filtering idiomatic *for the rest of time*). Plural-only forms also reject *of* [*a* N] (0 hits across tested complements), indicating the strict bare pseudo-partitive pattern rather than merely non-definite complementation. Full counts and methodology in Appendix A.2; raw data in `paper/data/coca-pilot.md`.

2.2 Collective NPs as a transitional case

Singular collective nouns (*committee, jury, family, team, government*; also *clergy, intelligentsia, public*, etc.) allow optional plural override of head-driven agreement, as in *CGEL* [8] (§18.2).

(4) a. *The committee is/are divided.*
b. *The committee consists/*consist of two staff and three students.*
c. *One committee, appointed last year, has/?have not yet met.*

The override in (4) is meaning-sensitive: singular agreement focuses on the unit, plural on the members. Plural is unavailable with predicates that apply to the whole but not individual members, and is "of questionable acceptability" if the collective takes determiner *one* or *another*.

Both collective NPs and number-transparent quantificational NPs override the head's morphological singular at agreement. The override is possible at all because plural agreement requires only non-singular construal, not the precise atomic enumeration that singular $a(n)$ and low cardinals demand. Individuation, the speaker's construal of referents as discrete units and the variable that count grammar tracks, comes in different intensities. Loose count properties (agreement, *many/few*, plural-only forms) can be satisfied by individuation alone; tight properties (singular $a(n)$, low cardinals) can't, since they require selecting exactly one atom or enumerating a precise n-tuple. That asymmetry is why plural agreement can override a morphologically singular head while tight properties on the same NP follow ordinary count-noun behaviour. The override is selective in this way: it operates at the loose end of the cluster, not across it. Traditional grammar calls the agreement-side reflex notional concord. Reynolds (2026a) develops the loose/tight asymmetry as a precision-driven dissociation pattern across English count nouns and proposes a bidirectional link between morphosyntax and individuation as the underlying mechanism, where each cues the other and the override is the case where individuation outruns the head's morphological signal.

The source of the non-singular construal differs in the two cases. In collectives, individuation is lexically encoded in the head: a singular collective denotes a unit whose members remain individuable, and the agreement relation accesses that member-level construal under predicates that license it. In number-transparent quantificational NPs, individuation is supplied by the *of*-complement: *a lot of tourists* sources non-singular construal from *tourists*; *a lot of money* sources low (non-count) construal from *money*; the construction projects the complement's count/non-count profile to agreement. The former is lexical-semantic accessibility of member individuation; the latter is constructional accessibility of the *of*-complement's individuation profile.

The two sources also differ in how stably the construal projects. The constructional source projects uniformly: in number-transparent QNs the override applies regardless of predicate type. The lexical-semantic source projects selectively: collective override requires the predicate to license member-level construal, which is why whole-unit predicates block the override even with the same singular collective head, and why singular-only determiners degrade plural override.

Both routes are mediated accessibility under the umbrella schema; the difference is whether the individuation profile is baked into the head's lexical semantics or imported from the *of*-complement.

Alongside *CGEL*'s override framing, the HPSG tradition encodes the same observation as index rather than concord agreement: concord tracks morphosyntactic features internal to the NP, while index agreement can track the referential or semantic index available to the agreement target

(Wechsler & Zlatić, 2003). That's a different formal encoding of the same accessibility relation, and it's consistent with §3's framework-neutrality.

CGEL itself notes that "the division between collective and number-transparent nouns is by no means sharply drawn" (§18.2). Some nouns straddle. *Couple* has two senses: *couple*$_1$ (a man and woman in a relationship) is a regular collective; *couple*$_2$ ('approximately two') is number-transparent. *Bunch* is the case where the profile framework does demotion work: *CGEL* claims it tilts collective for inanimate aggregates but transparent for groups of people, illustrated by the contrast in (5).

(5) a. *A bunch of flowers was presented to the teacher.*
 b. *A bunch of hooligans were seen leaving the premises.*

CGEL treats the first as collective and the second as transparent (§18.2). A pilot COCA check (May 2026) supports the animate half and gives no positive support for the inanimate half. Animate *bunch*-subjects are categorically plural (40 plural / 0 singular across *a bunch of people* and *a bunch of kids*), so the transparent reading holds. The inanimate-collective half gets no corpus support: the exact *CGEL* example *A bunch of flowers was presented to the teacher* returns 0 hits, inanimate *bunch* is largely absent from subject position (its typical use is as object, as in *she gave him a bunch of flowers*), and the two inanimate-subject hits that do surface take plural agreement rather than the predicted singular. Corpus absence of a constructed example isn't ungrammaticality, so this doesn't refute *CGEL*'s contrast as grammar; what it shows is that the inanimate-collective pattern doesn't project as a corpus regularity. The result is a localization of the projectibility claim: the *bunch* transparency profile is supported for animate aggregates, while the inanimate-collective contrast remains an illustrative intuition pending broader corpus or acceptability evidence. The schema's payoff here is a negative one, which is the point of having demotion conditions at all: it marks where a textbook contrast doesn't (yet) project. Full counts in Appendix A.4.

The schema accommodates both the paradigm and the transitional case, but they have different internal structure. Number-transparent NPs are class-like, with the override the default behaviour for class members. Collective NPs are gradient: the override probability depends on the head, the predicate, the determiner, the dialect register, and the construal the speaker has in mind. A theory of morphosyntactic transparency that ignored the difference between a class-based default override and a construal-driven optional one would lose the interesting structure of both cases.

2.3 Coordination as a non-instance

Coordination of singulars triggers plural agreement.

(6) *Bob and Jane are happy.*

The ordinary coordination in (6) looks like a transparency case at first (the conjuncts' number features percolate up to the coordinate construction and determine agreement), but the schema excludes it. The schema requires that X *could have* made P inaccessible. The coordinate construction does no such hiding: there's no head whose morphological number competes with the conjuncts' features for agreement.[4] The plural agreement is compositional addition (1 + 1 = 2), not mediation.

Some coordinate NPs take singular agreement.

[4]This verdict presupposes a headless (or numberless-head) analysis: with no conjunct serving as a number-bearing head, nothing conceals the conjuncts' summed features. The picture changes under an analysis on which one conjunct

(7) a. *Bacon and eggs is my favourite breakfast.*
b. *Wine and cheese is expected.*
c. *Fish and chips is expensive these days.*

Some of the same coordinate strings also take plural agreement.

(8) a. *Wine and cheese make the perfect party pair.*
b. *Fish and chips taste their best right after frying.*

The singular holistic cases in (7) and the plural counterparts in (8) leave the exclusion intact. In such cases agreement tracks a construal of the coordinate phrase: either a dish, package, event type, or conventional pairing, or a plural set of jointly mentioned items. The singular feature isn't a property of either conjunct that remains accessible through the coordinate construction; it's a construal of the coordinate phrase as a whole. Ordinary plural coordination and holistic singular coordination therefore differ in agreement semantics, but neither is transparent in the schema's sense. The former is compositional summation; the latter is whole-phrase unit construal.[5] *CGEL* treats coordination-and-agreement under a separate heading (§18.4), not alongside the override constructions of §18.2.

The clean distinction across §2's cases: in collective plural agreement, internal member plurality becomes accessible through a singular head; in number-transparent QNs, complement number becomes accessible through a quantificational head; in unit-construal coordination, the whole coordinate phrase is construed as singular and no internal property is being made transparent. The first two are mediated accessibility; the third isn't.

The "X could have made P inaccessible" clause works as an independent diagnostic, not an ad hoc filter. The test asks whether the construction-type admits a default-conceal variant: is there a version of the same construction-type, attested in the language, in which X's internal structure makes P inaccessible to R? For English NPs, head-driven agreement is the default and the number-transparent class is the marked exception within the same NP type. For coordinate constructions no such variant exists: the type admits no version in which conjunct features fail to reach agreement, so coordination has no default-conceal backdrop. The diagnostic distinguishes mediator-types from compositional-summation types: transparency is informative only where the same mediator type has opaque or partially opaque alternatives.

2.4 Number-transparent NPs as a projectibility profile

Section 1 set the diagnostic question: does identifying a candidate transparency profile let us predict behaviour? For number-transparent quantificational nouns the answer is yes, with the prediction running in two task settings.

For known *CGEL* members (*lot, plenty, lots, bags, heaps, loads, oodles, stacks, remainder, rest, number, couple*$_2$), lexical identity is sufficient input: knowing a noun is on this list licenses projection to the rest of the profile.

is the head. Closest-conjunct agreement (Bhatt & Walkow, 2013) is the case in point: where a verb tracks the nearest conjunct, that agreement conceals the resolved features and resolved agreement reaches them past it, so the same test can return an instance. As with the language-internal default-conceal baseline of §4, the verdict rides on the prior syntactic analysis.

[5] Cases like *One and one is two* are an arithmetic-expression construction, not a meal/package coordination: *and* functions close to an addition operator, and the subject denotes an operation, equation, or expression. Not transparency either.

For candidate new members, class discovery uses two diagnostics independently observable from any of the projected behaviours: quantificational semantics (an indefinite quantity rather than a precise count or kind) and high-frequency occurrence as head of an *of*-complement construction. A candidate that satisfies both is then tested against the projected cluster:

- near-obligatory agreement override in the central cases, singular- or plural-direction depending on subgroup;[6]
- characteristic determiner patterns by subgroup (*lot* most often with *a*, also with *the* in totality uses; *plenty* typically without a determiner; the plural-only forms typically without a determiner; *remainder, rest* typically with *the*; *number, couple*$_2$ typically with *a*);
- partitive vs. pseudo-partitive *of*-complement availability;
- behaviour under the fused-head construction;
- count/non-count percolation of the *of*-complement to the whole NP, with consequences for selection by predicates that are count- or non-count-sensitive;
- restricted premodification (*a whole/huge lot* but not *a remarkable plenty*).

The class's sub-group structure makes the cluster's predictions concrete. Table 1 displays the per-noun pattern for the central diagnostics; the entries are *CGEL*-coded rather than corpus-coded, and the per-noun corpus audit a fuller study would supply is the within-domain check that the falsification condition below leaves open.

Table 1. Number-transparent quantificational nouns: behavioural matrix (*CGEL*-based pilot).

Sub-group / Noun	Determiner	Partitive *of* N	Pseudo-partitive *of* N	Override direction
Singular, *a*-determined				
lot	*a*; also *the* (totality)	✓ (*a lot of the money*)	✓ (*a lot of money*)	sg → pl with plural complement
number	*a*	✓ (plural oblique)	✓ (plural oblique)	sg → pl with plural complement
couple$_2$	*a*	✓ (plural oblique)	✓ (plural oblique)	sg → pl with plural complement
Singular, usually without a determiner				
plenty	usually without a determiner	not typical	✓ (*plenty of money*)	sg → pl with plural complement
Plural, usually without a determiner				

[6] A pilot COCA check (May 2026) supports the near-categorical pattern: predicted-direction agreement runs 98–100% across nine tested QN+complement combinations after KWIC filtering of checked counter-direction cells. *A lot of money* required parse-shift filtering (10 of 13 raw plural-agreement hits were inside relative clauses modifying a plural head, e.g. *people who earn a lot of money are successful*; full filtering in Appendix A.3). The large *a lot of people is/was* counter-direction cell remains unfiltered, so the 98.0% row is conservative. The discriminating cells carry the evidence, and because filtering targeted counter-direction cells the figures corroborate the categorical pattern rather than estimate a rate. Full counts and methodology: Appendix A.3.

Sub-group / Noun	Determiner	Partitive *of* N	Pseudo-partitive *of* N	Override direction
lots, bags, heaps, loads, oodles, stacks	usually without a determiner	× (COCA pilot: ≈0.2%, §2.1)	✓	pl → sg with non-count complement
Singular, *the*-determined				
remainder	*the*	✓ (*the remainder of the time*)	× (**the remainder of time*)	sg/pl (complement-controlled)
rest	*the*	✓ (*the rest of the time*)	× (**the rest of time*)	sg/pl (complement-controlled)
Boundary				
majority, minority	*a*	✓	?	variable
bunch	*a*	✓	✓	variable

Override applies in the central cases for all class members; boundary cases show variable behaviour (§2.2). Fused-head behaviour and count/non-count percolation are predicted to apply across class members but aren't per-noun-audited here. Premodification is restricted across the class but with sub-group-specific patterns (*a whole/huge lot*; minimal modification with *plenty*; fewer attested premodifiers with the plural-only forms).

Each sub-group covaries: knowing a candidate noun's determiner pattern and partitive availability narrows the sub-group, which then licenses the override-direction prediction (and vice versa). The boundary cases have weaker covariance, which is why §2.2's gradient framing applies to them.

Figure 1 visualizes the override-direction data from the COCA pilot. The probative cells are the discriminating ones, where the head's number and the complement's diverge; there the override is categorical (plural with a singular-form quantificational noun and a plural complement, singular with a plural-form one and a non-count complement), and the few counter-direction tokens resolve on inspection into parse-shift and hypercorrection artefacts. The matched cells confirm the predicted direction but cannot tell the override apart from head-driven agreement, so they corroborate without testing. The check is exploratory: counter-direction cells were hand-filtered by KWIC while predicted-direction cells were not scrubbed to the same depth, so the proportions read as a categorical pattern with documented exceptions rather than as estimated rates, and the small-*n* discriminating cells (*plenty of money*, *the rest of the people*) carry wide uncertainty. Referential *the number of* is a negative control: the same noun agrees head-driven under *the* and overrides only under quantificational *a* (Appendix A.5), so the override belongs to the construction, not the lexeme.

This two-task structure keeps the prediction genuinely projective for the new-member case: inputs are quantificational semantics and *of*-complement-head distribution; the cluster of agreement-override / determiner-frame / partitive / fused-head / count-or-non-count / premodification behaviours is then testable on the candidate. Boundary cases like *majority* and *minority* (§2.2) and the dual-sense *couple*$_1$ (collective) versus *couple*$_2$ (number-transparent) show graded membership at the edges; the predictions hold in attenuated form, not as crisp class-conditional facts. What distinguishes prediction from redescription: the cluster supports inductive extension to *new* members (boundary

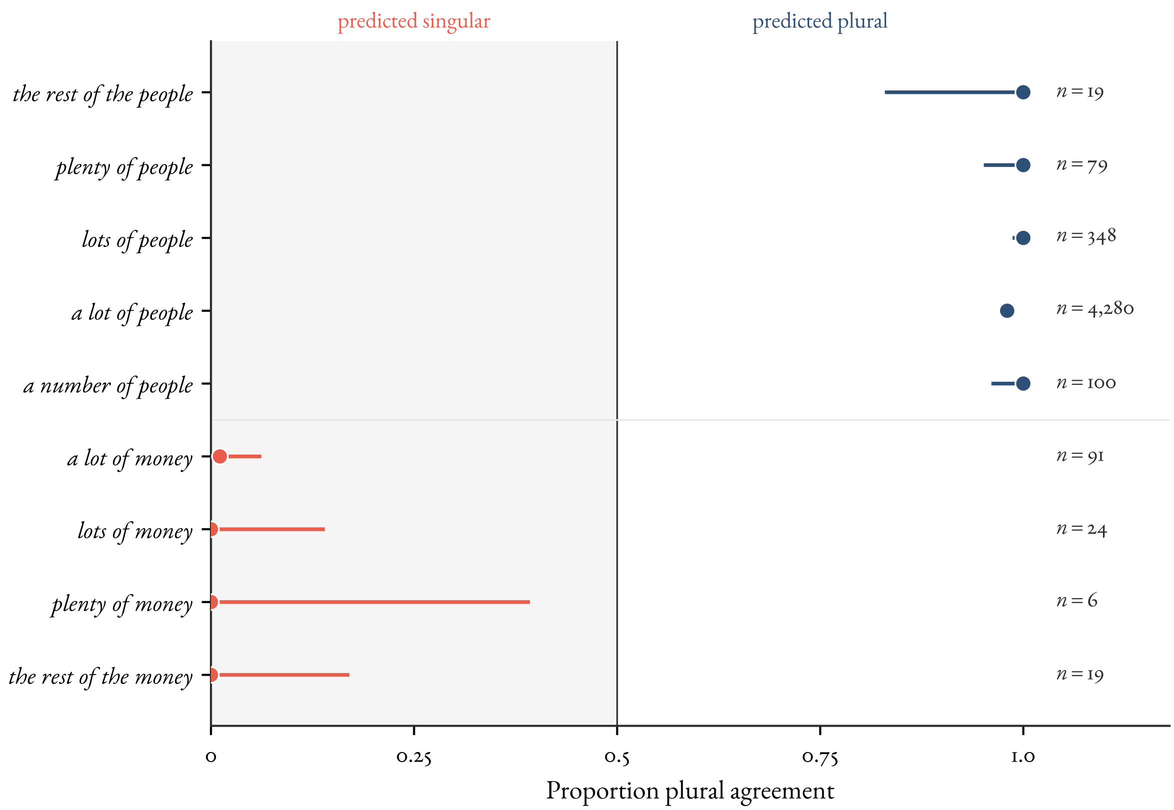

Figure 1: Override-direction agreement for nine number-transparent QN + complement combinations in COCA. Each row is a QN+complement; the x-axis is the proportion of finite-verb agreement in the plural direction (0 = all singular, 1 = all plural). Predicted-singular cases (coral) cluster at the left; predicted-plural cases (blue) cluster at the right. Horizontal bars are 95% Wilson binomial confidence intervals; *n* is the total finite-verb count for that subject after KWIC filtering where available. See footnote and `paper/data/coca-pilot.md`. Data: COCA pilot, May 2026.

cases like *majority/minority*; gradient cases like *bunch* in §2.2) that aren't built into the cluster's defining criteria.

The more demanding test is what the profile lets us predict beyond the in-grammar cluster. Three off-list projections are plausible candidates. *Diachronic*: members should enter the class through grammaticalization or sense-split rather than through arbitrary lexical addition, with recent boundary additions (*majority, minority*) and the sense-split in *couple* matching that pattern. *Processing*: if the override is grammaticalized rather than improvised, sentences with number-transparent QNs should carry no comprehension cost relative to ordinary plural-subject sentences in adult speakers; no-cost or low-cost override is the prediction, not a parsing penalty for the surface mismatch. *Cross-linguistic*: the comparative concept of mediated accessibility (§4) licenses asking whether languages with analogous partitive-quantificational configurations exhibit comparable cluster structure (French *beaucoup de*, Spanish *un montón de*); the descriptive categories needn't be equated for the cluster shape to be compared. None of these is tested in this paper. Their availability as predictions, rather than restatements of the input criteria, is what supports treating the class as a projectibility profile in §5.1's stronger sense: a maintained-and-projectible cluster rather than just a distributional grouping.

The cluster of behaviours listed above makes number-transparent quantificational nouns the strongest candidate PROJECTIBILITY PROFILE examined here: an observable base (lexical class membership for known members; quantificational semantics plus *of*-complement distribution for candidates) supporting projection to a structured set of targets, stabilized by conventional lexical inheritance. The profile is a candidate to pass Slater's (2015) stability-plus-projection diagnostic; the within-domain check beyond override direction is empirically open. The falsification condition is concrete: if a noun satisfying the input diagnostics failed to show the predicted cluster, the profile would fail. Boundary cases (*majority*, *minority*, sense-split *couple*) reflect the conventional inheritance that stabilizes the profile. The count/non-count percolation behaviour the profile licenses participates in the broader count cluster that Reynolds (2026a) analyzes as Khalidian, with conventional individuation as a central causal node; §5.4 returns to it. Whether the profile earns the local-kind label as a downstream verdict is for §5.

2.5 Transparent free relatives

A second morphosyntactic-transparency case fits the schema. Transparent free relatives (TFRs) are a sub-type of fused-relative construction (*CGEL*, ch. 12, §6.3) in which the matrix accesses properties of an embedded predicate (the NUCLEUS) through *what*: *what they call ecstatic*, *what we regard as essential*, *what seems to be a tourist*. This paper uses TFRs only as a second profile case, following Kim's (2026, ch. 6.8) construction-grammar treatment and the joint working note (2026).

(9) a. *What appear to be trousers are on the chair.*
b. *What they call essential workers are exhausted.*
c. *He was what they call dimwitted.*

For present purposes, the profile claim is enough. TFRs support two *R*-relations: agreement transparency, where matrix agreement accesses the nucleus's number profile through singular *what*, and syntactic-category transparency, where the matrix accesses the nucleus's category through the *what*-clause. Example (9c) illustrates the AP-nucleus side; the working-note data add *what I consider important* and *what he takes to be obvious*.

The observable base is constructional sub-type plus verb-class membership; the projected targets are parent-level agreement transparency and daughter-level syntactic-category transparency; the stabilizer is constructional inheritance. The verb-class corpus detail, AP/PP-nucleus stratification, and analysis of the attributional licence are deferred to Kim and Reynolds (2026).

2.6 Korean as the cross-linguistic test

Korean is the comparative concept's most consequential cross-linguistic test, under joint development rather than reported here. A Korean case-stacking or scrambling analysis binds X to a layered case or word-order configuration, P to an argument-role, scope, or information-structural property, and R to role recovery, scope interpretation, or discourse licensing. Those bindings need Korean-particular diagnostics, so this paper sets up the test without running it: the Korean answers are forthcoming co-authored work, not evidence for the English profiles established here. §4 returns to this as a methodological constraint and §5 as the umbrella's central open test.

2.7 Bridge

The cases in Section 2 share a relation: morphosyntactic matching mediated by a construction or lexical class. Section 3 shifts the relation, taking interpretation, rated transparency, and processing as R instead of agreement; the form–meaning arm is developed in separate work.

3 Form–meaning transparency

Section 2's relation was agreement. The schema isn't confined to it. Where R is interpretation, learning, or processing rather than agreement, the same question (does P remain accessible through X when X might have concealed it?) reaches a second family: the form–meaning transparency of constructions, compounds, and morphologically complex words. The relation is broader and the accessibility is gradient rather than class-like, but the slots bind as before. This section makes the case briefly; the form–meaning family is developed in its own right in separate work.

Three windows make the point. Construction Grammar treats constructions as learned form–meaning pairings (Goldberg, 2006), and their compositional transparency runs a gradient: from fully compositional patterns, through partly schematic and idiomatically constrained constructions, to frozen idioms like *kick the bucket*, where the parts' meanings no longer reach interpretation through the whole. The frozen idiom isn't an exclusion but the trivial limit of a real gradient: the construction-type admits a concealing variant, so it sits on the scale rather than off it (contrast coordination, §2.3, whose type admits no concealing variant at all). A plausible strengthening is that transparency is partly a *network* property, a construction's compositional reading being accessible partly because related constructions in the constructicon support it (Kay & Fillmore, 1999); that is a hypothesis, not a result.

Two quantitative windows give the gradient empirical content. Bell and Schäfer (2016) model the rated transparency of English noun-noun compounds from corpus-statistical predictors (constituent frequency, semantic-relation expectedness, head productivity), so a compound's parts are accessible to a degree that tracks its position in the network of compounds. Heyer and Kornishova (2018) find that rated transparency modulates masked morphological priming, but only at long stimulus-onset asynchronies: an early, semantically blind decomposition stage gives way to a later transparency-sensitive one. (Feldman and colleagues (2015) contest the serial reading; the dispute is live.)

These are three within-domain profiles, distinct in items, methods, and stabilizers (constructional inheritance, usage entrenchment, processing architecture), and each licenses within-domain projection. Whether they're joint reflexes of a single underlying organization (a genuine cross-domain cluster) or three separate profiles is an open empirical question that a within-items probe would settle; that probe and the family's fuller development are taken up in separate work. Here the windows do one job: they show the schema's R isn't fixed. A construction can be transparent for one relation and opaque for another. *Kick the bucket* is opaque for interpretation yet morphosyntactically regular, its verb agreeing with its subject in the ordinary way; the number-transparent NPs of §2 are constructions too, transparent for agreement whatever their interpretive compositionality. Accessibility isn't transitive across R-values. That R-relativity is built into the umbrella, and it's a feature: the schema asks whether P remains accessible *for the purposes of* a specified R, not in general. The umbrella analysis is also framework-neutral here, since *CGEL*, Sign-Based Construction Grammar, and Goldbergian CxG encode the override differently and the schema sits above the choice.

3.1 Referential transparency as the schema's outer boundary

One case sits at the schema's edge.

(10) a. *Lois Lane believes Superman is brave.*
b. *Lois Lane believes Clark Kent is brave.*

In an intensional context, substituting *Clark Kent* for *Superman* in (10) may not preserve truth-conditions: the *de re* reading allows the substitution, the *de dicto* reading doesn't. The schema technically applies, with X the intensional context, P the referent of the embedded singular term, and R substitution under co-reference. What makes it the boundary case is that X is a *context*, not a *form* whose internal structure could have mediated access to P. The schema stretches thinnest here, and the stabilizing source is the formal-semantic structure of the embedding rather than anything constructional. The philosophical literature on referential transparency (Quine (1956) as the canonical anchor) would dominate the case if developed; here it marks a real heterogeneity in the umbrella, handled by the schema's R-relativity rather than forced into the form–meaning family.

4 Typological transparency

This section is methodological. The previous sections worked through cases in English and Russian; Section 2.6 names Korean case-stacking and scrambling as a deferred extension rather than as evidence in this paper. How should the paper handle the move from those language-particular cases to any cross-linguistic claim? Haspelmath's (2010) descriptive-versus-comparative-concept distinction is the right discipline.

4.1 Haspelmath's distinction

Haspelmath (2010) argues that crosslinguistic comparison shouldn't assume universally available descriptive categories[7] like ADJECTIVE, PASSIVE, ACCUSATIVE, or SUBJECT. Each language has its own descriptive categories, defined by the language-particular distributional and morphosyntactic

[7] Throughout, *category* is used in Haspelmath's *descriptive category* sense: a language-particular grammatical grouping fixed by a language's own distributional criteria, and neutral on whether that grouping is projectible. This differs from the sense used elsewhere in this research programme, on which a *category* just is a projectible kind (Reynolds, 2026b); that kind-level notion is carried here by the *projectibility profile* (§5.1), not by *category*.

criteria that make the categories work in that language (Haspelmath, 2010, p. 664). The criteria differ from language to language: what counts as a passive in one language needn't be what counts as a passive in another, so descriptive categories can't be equated across languages.

Comparative concepts, by contrast, are analyst-created tools designed for the purpose of crosslinguistic comparison. They aren't part of any language system and aren't psychologically real; they can only be more or less well suited to the comparative task, rather than right or wrong (Haspelmath, 2010, p. 665). Comparative concepts have to be universally applicable, which means they have to be built from universally applicable building blocks: conceptual-semantic concepts (such as notions of agency or possession), general formal concepts (such as "phonological word"), and other comparative concepts.

Haspelmath calls the alternative position categorial universalism: the assumption that there's a substantial set of crosslinguistic descriptive categories from which languages select (Haspelmath, 2010, p. 663). Newmeyer's (2007) *Linguistic typology requires crosslinguistic formal categories* is the canonical statement of the position Haspelmath rejects: typological generalizations, on Newmeyer's argument, can't be made through purely conceptual-semantic comparative concepts because such generalizations depend on reference to the specific form in which the concepts are realized (Newmeyer, 2007, p. 136). Haspelmath (2010, p. 666) grants the realization point but argues that the relevant formal references are themselves comparative concepts (e.g. "adposition", "zero coding") definable in conceptual-semantic and general-formal terms, not crosslinguistic descriptive categories. Categorial particularism, his preferred position, holds that descriptive categories are language-particular and that comparison happens through analyst-defined comparative concepts that stand in many-to-many relations with descriptive categories. The position has been developed further in typological work.

The methodological consequence is straightforward. A typological generalization isn't a claim about a descriptive category instantiated in every language. It's a claim about how analyst-defined comparative concepts are realized in language-particular ways. The generalization that nominative case-marking is widespread across languages doesn't entail that "nominative" picks out the same descriptive category in any two languages.

4.2 The umbrella schema as a comparative concept

The umbrella schema introduced in §1 is a comparative concept. It's built from conceptual-semantic notions (mediation, accessibility, property, relation), plus relational variables (X, P, R) whose values are filled in language-particularly. The schema's question (X *is transparent w.r.t.* P *for* R *when* P*, which* X *might otherwise have concealed from* R*, remains accessible through* X *for the purposes of* R) is something the analyst asks of any language; it's not a claim that any particular language has a descriptive category corresponding to TRANSPARENCY.

The schema's primitives meet Haspelmath's (2010, p. 665) standard for comparative concepts (conceptual-semantic notions, general formal notions, other comparative concepts). *Property* (P) and *relation* (R) are conceptual-semantic primitives. *Form* (X) is a general formal notion. *Accessible through* is the contested slot. Accessibility has discipline-specific senses (discourse-pragmatic salience, psycholinguistic retrievability, formal-semantic substitution-licensing); the schema's sense is more abstract: P is accessible to R through X when R tracks P across X rather than tracking some property of X itself. The notion is conceptual-semantic, not psycholinguistic. The case studies operationalize it differently (agreement targeting, constructional interpretability, rated transparency, priming magnitude), but the comparative concept lives at the abstract level. *Mediation* is the schema's other central notion, tested by the contrast between paradigm cases and the §2.3 non-instance.

The non-instance test (§2.3) carries a language-internal baseline, and naming it as such is what keeps the comparative concept honest. The test asks whether the same construction-type has an attested default-conceal variant in the language under study: a version of X whose internal structure would make P inaccessible to R. For English number-transparent NPs that baseline is head-driven agreement, and the transparent class is the marked exception against it; the baseline is not exported from English. The decision procedure is the same in any language: identify X, P, and R by that language's own distributional criteria, then ask whether the language attests a same-type variant in which P is concealed from R. Where it does, the transparent case is informative; where the construction-type admits no concealing variant (coordination in English, §2.3), the schema returns a non-instance. Because the baseline is fixed language-internally, *accessible through* imports no English-particular grammar: it inherits whatever default the local construction-type supplies.

Each value of the schema's variables is language-particular. In English, X is the number-transparent quantificational noun + *of*-complement construction (a *CGEL* descriptive category); P is the number/count profile of the complement; R is subject-verb agreement. In the Korean extension named in Section 2.6, X may be a layered case-marking pattern or a scrambled word order; P may be argument-role assignment, scope, or information-structural status; R may be argument-role recovery, scope interpretation, or discourse licensing. The variable bindings are different. The comparative concept MEDIATED ACCESSIBILITY is the same.

Haspelmath's framework requires exactly this: cross-linguistic claims are made through comparative concepts whose realizations differ from language to language. The English and Korean cases aren't "instances of the same descriptive category." They're different descriptive categories that can be compared through the comparative concept of mediated accessibility, with different X, P, and R values.

The form–meaning accessibility family (§3) isn't itself a comparative concept. The schema (mediated accessibility, with its $X/P/R$ variables) is the comparative concept; the family is a set of language-particular phenomena and theoretical/experimental measurements compared *through* the schema. CxG constructional transparency, compound rated transparency, and morphological priming are different operational windows on form–meaning accessibility (worked through in English with one Russian extension): a theoretical profile, a measurement, and an experimental effect. They're intended to bear on the same comparative-concept gradient. The cross-linguistic move at the comparative-concept level doesn't commit the paper to crosslinguistic descriptive categories.

4.3 Worked example: descriptive category and comparative concept

Consider the typological question: do all languages have number-transparent constructions? The categorial-universalist answer is to look for English-style number-transparent NPs in every language. The categorial-particularist answer is different. NUMBER-TRANSPARENT QUANTIFICATIONAL NOUNS is a *CGEL* English descriptive category, defined by the override behaviour (near-obligatory in the central cases), the small and relatively stable membership, the characteristic determiner patterns, and the partitive/pseudo-partitive *of*-complement restrictions worked through in §2.1. Crosslinguistic comparison tracks the comparative concept *organizing* that descriptive category, not the descriptive category itself.

What the comparative concept of mediated accessibility lets us ask is different. Across languages, which configurations let a property remain accessible to a grammatical relation that the surface form might otherwise have hidden? English number-transparent NPs answer the question for the relation R = subject-verb agreement, with P = number/count profile of the *of*-complement. The comparative concept doesn't predict that any other language has matching descriptive categories. It predicts that the analyst's question is askable of any language, and that the answers will differ.

The cross-linguistic extensions the paper has on hand are limited. The deadjectival-nominalization parallel in English and Russian noted in §3 is one: two languages, two descriptive categories (deadjectival nominalization in each), but the same comparative-concept question asked of both, with parallel answers. The Korean extension in Section 2.6 is the most consequential next test, deferred for separate co-authored development. Beyond English, the same question is taken up below for Spanish and, in a different marking type, Tsez; even so, the cross-linguistic claim at the comparative-concept level is a structural one (the comparative concept lives above the descriptive category, whatever the language) rather than a fully demonstrated comparative result, and Korean remains the deeper prospective test.

The claim sits at the right crosslinguistic level. It requires neither a universal descriptive category TRANSPARENCY or NUMBER-TRANSPARENT nor a claim that any two languages have "instances of the same construction." It claims, more modestly, that the comparative concept of mediated accessibility can be applied to any language and that the answers it elicits are language-particular categorial content. That modesty is what categorial particularism asks of comparative work.

4.4 Cross-linguistic cases and topologies

Two worked languages let the question be asked outside English, one of them outside the agreement type English uses. Spanish has the number-transparent profile itself. In a partitive subject like *la mayoría de los manifestantes* ('the majority of the demonstrators'), the finite verb may agree in the plural with the complement (*gritaban* 'were shouting') or in the singular with the quantificational noun *mayoría* (*gritaba*); the plural is the more frequent option and the only one available under a plural predicative (Real Academia Española, n.d.). Both the concealing variant (singular, agreeing with *mayoría*) and the transparent variant (plural, agreeing with the complement) are attested, just as for English *a number of* versus *the number of*. The profile travels, with the same alternation, to another language.

Tsez (Nakh-Daghestanian; head-final and ergative; about seven thousand speakers) gives a case in a different marking type. Tsez verbs prefix-agree in noun class with their absolutive argument, so the agreement is marked on the verbal head. A clausal argument is class IV, the class that includes clauses, so a matrix verb's default is class IV agreement with the complement clause as a unit; but when the embedded absolutive is a topic, the matrix verb can instead agree with it (Potsdam & Polinsky, 1999, p. 436).

(11) a. enir [užā magalu b-āc'ru-łi] r-iyxo
mother [boy-ERG bread.III.ABS III-ate-NMLZ] IV-knows
'The mother knows the boy ate the bread.' (default: the matrix verb is class IV, agreeing with the clause)

b. enir [užā magalu b-āc'ru-łi] b-iyxo
mother [boy-ERG bread.III.ABS III-ate-NMLZ] III-knows
'The mother knows the boy ate the bread.' (long-distance agreement: the matrix verb is

class III, agreeing with embedded *magalu* 'bread')

In the schema's terms, X is the embedded clause, P is the noun class of the embedded absolutive, and R is matrix verb agreement. The default-conceal variant is the class IV agreement in (11a), attested in the same language and the same sentence, and (11b) is the transparent override. The baseline is not "the nearest head's own feature", as with English number-transparent nouns, but "default class IV for a clausal argument": a different default in a different language type. That is what the comparative concept predicts, the question constant and the language-particular answer not. Whether the override is analysed as covert movement or as long-distance agreement is a framework-internal question the schema brackets, as with extraction in §2; the fuller analysis is in Polinsky and Potsdam (2001).

Spanish and Tsez are both outward endpoint cases: a property associated with material inside X reaches an external agreement relation. The same schema maps several other accessibility topologies, provided each remains subject to the §2.3 default-conceal test: identify X, P, and R, then ask what same-language pattern would make the proposed access disappear.

Table 2. Accessibility topologies under the umbrella schema.

Topology	X–P–R binding	Falsification condition and status
Outward endpoint access	X = NP or clause mediator; P = complement or embedded feature; R = external agreement. English QNs and Tsez long-distance agreement instantiate this topology.	Fails where only the mediator's own head/class feature controls R, or where the construction-type has no concealing variant. This is the paper's core positive pattern.
Null-source subpart access	X = predicate configuration containing a possessed NP with an unpronounced possessor; P = the recoverable possessor's person/number; R = anaphoric verbal morphology plus subject case. Aleut is the key case (Merchant, 2011, p. 195).	Fails if the omitted possessor patterns like an ordinary overt possessor, or if the anaphoric verbal suffix appears without the missing subpart. Positive with a qualification: Merchant analyzes the missing possessor as syntactically present pro.
Inward dependent access	X = clause or nominal-dependency domain; P = clausal TAM/case value; R = morphology on nominal dependents. Lardil and Kayardild show clausal TAM values on dependent nominals (Nordlinger & Sadler, 2004).	Fails if dependent marking is only local nominal morphology, with no covariation with clausal predication. This is an observational topology; no generative analysis is needed for the comparison.

Topology	X–P–R binding	Falsification condition and status
Path-distributed access	X = extraction path; P = grammatical relation/case of the extracted item; R = wh-agreement morphology along that path. Chamorro supplies the comparison (Chung, 1998, 2004).	Fails if the morphology appears only at the gap or endpoint, or if it doesn't vary with the extracted item's role. Positive as a topology, but extraction remains outside this paper's worked evidence.
Feature relay boundary case	X = possessed NP; P = possessor obviation/proximation; R = possessed-noun morphology and verbal obviation agreement. Kutenai and Algonquian obviation are useful near misses (Dryer, 1992).	Fails as transparency where the predicate sees only a feature already relayed onto the possessed head. Boundary case: transformed access, outside the direct-mediated pattern.

Aleut deserves separate mention because it changes the earlier head-marked caution. Merchant's missing-possessor contrast shows that when the possessor is overt, Aleut has ordinary agreement; when that possessor is unpronounced, the subject appears in relative case and the verb takes an anaphoric suffix (Merchant, 2011, p. 195).

The controller is recoverable as a subpart of a non-subject NP, but it needn't be overt. In the schema's terms, P can be null-source, provided the language supplies an independent contrast showing that R tracks the missing subpart rather than the whole NP.

This is a positive case with a qualification. It differs from the simple overt-head-marking analogue of English QNs because Merchant's own analysis treats the missing argument as syntactically present pro moving into a local relation with T. The comparison therefore sharpens the typology rather than overturning the diagnostic: overt intact-NP transparency and null/anaphoric subpart accessibility are different topologies.

A targeted check of head-marked systems adds the remaining caution. If English-style NP transparency were easy to replicate in head-marked grammar, the clearest cases should involve verbal indexing by an overt embedded nominal inside a larger NP while that nominal remains internal. The available positive candidates instead cluster around possessors, and most require a further analysis.

Chimane is the strongest near case: an apparently NP-internal possessor can control object agreement on the verb, but Ritchie's analyses treat the relation as mediated by applicative-like morphology and a clause-level proxy of the internal possessor (Ritchie, 2014, 2016). Other apparent positives are explicitly externalized: Nez Perce object possessor agreement is possessor raising, with the possessor moving from a possessum-DP-internal position to a clausal A-position (Deal, 2013); Deal's broader survey treats many comparable cases as external possession, possessor raising, incorporation, or related clause-level dependencies rather than as agreement with an embedded dependent inside an intact NP (Deal, 2017).

Yucatec Maya gives the useful negative backdrop: it is an exclusively head-marking language (Bohnemeyer et al., 2015, p. 51), but in inalienable-possession predications the grammar consistently keeps possessor and possessum inside possessed NPs, apart from the existential possessor construction (Lehmann, 2002, pp. 113, 116, 127). The diagnostic point is narrow: overt sensitivity to a possessor is not enough. The decisive question is whether an embedded dependent remains NP-internal while controlling clause-level morphology.

4.5 A potential objection and a reply

A reader might object that the umbrella schema looks like a universal descriptive category smuggled in as a comparative concept. The sharpest form of the objection targets the non-instance test: *accessible through* presupposes a default-conceal baseline, and for the English cases that baseline is head-driven agreement, a language-particular fact. If the schema's deepest primitive is parasitic on an English-shaped default, then calling it universally applicable looks like categorial universalism in disguise.

The reply isn't that the schema is "just a question": a universal category can be framed as a question too, so that move wouldn't tell a comparative concept apart from a category in disguise. The reply is that the baseline the objection points to is fixed *language-internally*, which is how comparative concepts are meant to work. The default-conceal variant is not assumed to be English head-driven agreement; it's whatever same-type concealing variant the language under study attests, by the procedure set out earlier in this section. So the schema doesn't import an English default; it inherits each language's own.

Three things follow that a universal descriptive category wouldn't deliver. The variables are bound by language-particular distributional criteria, not by a fixed cross-linguistic definition. The schema stands in many-to-many relations with descriptive categories: one language's category can fill X, P, or R under different bindings, and several categories can realize one binding. And it carries explicit falsification conditions: a candidate is a non-instance where the construction-type admits no concealing variant, and a transparency attribution is demoted where the predicted access fails to project, as in the *bunch* localization of §2.2. The honest residue is that the test is only as sharp as the prior individuation of construction-type: deciding what counts as the same type with and without concealment is itself a language-internal analysis, so the diagnostic is crisp where that individuation is independently motivated and softer where it isn't. Korean (§2.6) is the test of whether the procedure runs on a baseline that isn't agreement at all. Haspelmath's (2010, p. 665) criterion for a good comparative concept, that it can be more or less well suited to the task of permitting crosslinguistic comparison, is exactly the standard the umbrella schema is meant to meet.

Newmeyer's (2007) specific argument deserves a direct response. His central case is that typological generalizations must reference not just a universal concept but the specific form in which it's realized: negative particles and negative auxiliaries express the same concept of negation, yet only the latter pattern with verbs in word-order typology, so a generalization stated over the concept "negation" misses the regularity, and capturing it requires equating formal constructs such as "auxiliary" across languages (Newmeyer, 2007, p. 136). The point is correct on its own terms: concept-reference alone is insufficient for such generalizations. But it doesn't force crosslinguistic *descriptive* categories.

The formal reference Newmeyer demands enters the schema at the X slot, and X is a general-formal comparative concept in Haspelmath's (2010, p. 666) sense (like "adposition" or "zero coding"), fixed by each language's own criteria rather than equated token-for-token. The English X is the number-transparent quantificational noun + *of*-complement construction; the English TFR X is the fused-relative construction with attributional-verb licensing; the Korean X (Section 2.6) is whatever Korean construction the analyst identifies. The default-conceal baseline that gives the non-instance test its grip is itself such a general-formal comparative concept, not an imported universal category. Newmeyer's deeper worry, that thoroughgoing particularism leaves us unable to say two speakers share a category at all (his Section 5), bears on radical categorial particularism in general; the schema doesn't deny that languages can share general-formal comparative concepts, it only declines to presuppose a fixed universal inventory of descriptive ones.

The metaphysical conclusion in §5, that the umbrella schema is a tool for locating projectibility profiles rather than itself a kind, is consistent with this. Three levels stay distinct: the comparative concept (mediated accessibility) lives at the cross-linguistic methodological level; descriptive categories (English number-transparent quantificational nouns, Korean case-stacking and scrambling, and so on) live at the language-particular grammatical level; PROJECTIBILITY PROFILES are observable bases licensing non-trivial projection, stabilized by identifiable sources, realized within or across such descriptive categories. Kindhood is a downstream verdict on profiles that earn it. The paper doesn't collapse the levels.

4.6 Methodological default

Haspelmath's distinction has been contested in the typological literature; this paper doesn't adjudicate. Haspelmath's position is taken as the safer methodological default for cross-linguistic claims, since it requires no commitment to descriptive-category universality and doesn't assume a substantive crosslinguistic ontology of grammatical categories. Readers more comfortable with categorial universalism can read the umbrella schema as a useful working concept whose status is secondary to its methodological pay-off.

Croft's (2001) Radical Construction Grammar (RCG) is the construction-grammar variant closest to the categorial-particularist commitment adopted here: it routes cross-linguistic comparison through language-particular constructions linked by similarity rather than through universal categories or formal apparatus, and the schema's variables (X, P, R) can be instantiated by language-particular constructions in just that way. The paper routes through Haspelmath-style comparative concepts rather than RCG's construction-similarity hierarchies because the schema is meant to sit above any single construction-grammar variant (*CGEL*, SBCG, Goldbergian CxG, or RCG), and because the RCG route is not obviously freer of crosslinguistic commitments: Newmeyer (2007, fn. 22) argues that RCG's constructions implicitly entail crosslinguistic formal categories of just the kind they are meant to avoid. Developing the comparison through RCG is left for further work; both routes preserve the categorial-particularist commitment.

4.7 Bridge to §5

Section 5 turns to the metaphysics: which transparency profiles are stabilized well enough to project, and how the three levels relate. One point can be made now: cross-linguistic projection at the profile level isn't entailed by cross-linguistic projection at the comparative-concept level.

5 Projectibility profiles

Section 1 set the paper's central question: when does a transparency attribution support induction, and what stabilizes that induction? In Boydian terms, the question is field-relative: a profile is projectible for a field's explanatory and inductive tasks (Boyd, 1991, 1999). The primary field here is syntax. The strongest profiles support induction over agreement, syntactic-category licensing, complement selection, and related grammatical behaviour.

This section collects the answers and keeps the field-relative claims separate. The umbrella schema is a tool for locating projectibility profiles, not itself a profile, and the profiles the paper has examined are stabilized by heterogeneous structures.

5.1 The umbrella isn't a single profile

Section 4 established that the umbrella schema is a comparative concept, an analyst's tool for asking the projectibility question of any language. The metaphysical question is what *follows* from posing the question across a heterogeneous family of cases. The umbrella's heterogeneity is the answer. The X-values across the family include lexical classes (number-transparent quantificational nouns), constructions (the ditransitive, partly-filled idioms), compounds (NN compounds), derivationally complex words (*-ness* and *-ost'* nominalizations), and intensional contexts (referential transparency). The P-values include number/count features, constituent meanings and their relations, compositional contributions, lexical-stem identity, and referents. The R-values include subject-verb agreement, interpretation, masked priming, and substitution under co-reference. No single stabilizing source unifies these triples, and no single field's inductive base supports projection across them.

The umbrella identifies many candidate profiles, not one. A transparency label, on its own, licenses no projection: knowing only that something counts as *transparent*, that the schema's X-P-R slots are filled, lets us predict nothing substantive about it. The projectible content lives at the level of the individual profile, not at the level of the schema or the label. The schema's role is to make the projectibility question askable, case by case. That role isn't empty. The schema contributes the relational discipline of §2.3, the non-instance requirement that X could have made P inaccessible to R, and that requirement does analytical work no per-profile description supplies on its own: it excludes coordination as compositional summation (§2.3), places frozen idioms at the trivial limit of a real gradient rather than off it (§3), and marks where a *transparency* label picks out a genuine mediator rather than restating a distribution. The umbrella isn't a kind; its contribution is this relational discipline.

A useful general distinction follows from this heterogeneity. A *distributional class* names a real grouping at the surface (the schema's $X/P/R$ slots get filled by some real configuration), while a *projectibility profile* also requires some identifiable source of stability keeping the cluster projectible (a stabilizer in the sense of §5.3, not necessarily a causal mechanism). The umbrella schema names a distributional class but not a projectibility profile: the schema picks out a real family of cases, but no single stabilizing source unifies them at the umbrella level. Each profile examined below has to be evaluated separately for whether it qualifies as projectible in this stronger sense. The negative-polarity-item literature offers a parallel pattern: NPIs share a surface licensing property but the licensing patterns Hoeksema (2012) catalogues aren't manifestations of one stabilizing source, so NPIs are a class but not a projectibility profile at the umbrella level (developed at length in (Reynolds, 2026b, ch. 7)).

5.2 Syntactic and adjacent profiles

The paper's primary profiles are syntactic.

Number-transparent quantificational nouns (§2.4). The observable base is class membership, identified by lexical identity, quantificational semantics, corpus distribution as head of an *of*-complement construction, and typical determiner frame. The projected targets are non-trivial: near-obligatory agreement override in the central cases (in either direction); characteristic determiner patterns by subgroup; partitive vs. pseudo-partitive *of*-complement availability; behaviour under fused-head construction; count/non-count percolation of the *of*-complement; restricted premodification. Class membership generates expectations for *new* members (boundary cases like *majority* and *minority*; gradient cases like *bunch*) that aren't built into the class's defining criteria. This is the strongest candidate profile examined here: the cluster supports inductive extension, not just cataloguing of the enumerated members.

Transparent free relatives (§2.5). The observable base is constructional sub-type and verb-class membership. The TFR parent licenses agreement transparency across the full verb inventory; the attributional daughter licenses syntactic-category transparency as well. Object-oriented attributional verbs (*call, consider, describe as, regard as, take to be*) predict AP-nucleus productivity and the narrower syntactic-category-transparency effect; the *seem/appear* class predicts agreement transparency without syntactic-category transparency. The corpus check is still working evidence, but the profile is syntactic in the same field-relative sense as number-transparent QNs: a constructional input projects to grammatical behaviour.

The adjacent form–meaning profiles belong to neighbouring fields rather than to syntax itself. The form–meaning accessibility family (§3) contains three within-domain profiles, related by analytical theme rather than by demonstrated cross-domain projection. First, a construction-level profile: graded compositionality predicts compositional interpretability via inheritance and network position. Second, a compound-level profile: corpus-statistical predictors predict rated transparency. Third, a derivationally-complex-word-level profile: rated transparency predicts long-SOA priming magnitude. The construction-level profile is theoretical; the other two rest on direct empirical evidence (Bell & Schäfer's models; Heyer & Kornishova's experiments). Whether the three compose into a cross-domain cluster on shared items is an open empirical question; on present evidence they don't, and a within-items probe (corpus predictors, ratings, priming, interpretation accuracy on the same items), designed in separate work, would settle it. The form–meaning family is an open research programme pursued in separate work, not a delivered result of this paper.

Acquisition trajectory is a plausible further extension consistent with the network claim of §3 but not directly grounded by these sections; sourcing usage-based acquisition evidence is left for further work.

5.3 Stabilizing structures

Different structures stabilize the profiles, and the paper's R-values across §§2–3 (plus the boundary case of §3.1) call for a small inventory of stabilizing sources rather than a single mechanism.

Lexical convention stabilizes the number-transparent nouns as a class. Lexical inheritance here means that speakers acquire not isolated words but item-specific constructional slots: *a lot of* X, *plenty of* X, *the rest of* X, *a number of* X, with their associated agreement and complement-selection behaviour. The acquisition target is the slot, not just the lexeme. Speakers acquire a small, relatively stable inventory of these slots; new members enter at the boundary (*majority, minority*) or by sense-split (*couple*$_2$). Class identity itself isn't held by Boyd-style homeostatic feedback; it has the stability characteristic of any small, conventionally established grammatical class. The count/non-count percolation behaviour the class licenses connects to broader patterns in English countability; (Reynolds, 2026a) develops a Khalidian causal-cluster analysis of those patterns with conventional individuation as a central node. The transparency paper draws on that analysis but doesn't depend on its stronger claims; the within-NP percolation can be observed and predicted independently.

Constructional inheritance stabilizes the TFR daughter profile (§2.5) and the network-relative transparency hypothesis of §3. In TFRs, the parent and attributional daughter support different access conditions. In the form–meaning family, a construction's compositional accessibility depends partly on the inheritance hierarchy it sits in; constructions whose inherited supercategories are well-attested support stronger compositional projection than orphaned constructions.

Usage-based entrenchment stabilizes the corpus-statistical predictors of compound transparency (§3). Bell and Schäfer's frequency, productivity, and semantic-relation expectedness predictors are corpus-derivable measurements of usage entrenchment; the rated-transparency target is a steady-state property whose stability is plausibly a reflex of that entrenchment.

Processing architecture stabilizes the time-course modulation of priming (§3). Heyer and Kornishova's SOA-dependent transparency effect is a processing-architecture observation: the time-course of morphological decomposition makes morpho-orthographic information available before morpho-semantic information, and the resulting priming pattern depends on representational properties that the processing architecture exposes only at certain stages.

Formal-semantic environment stabilizes the referential transparency cases (§3.1). Substitution-salva-veritate behaviour depends on the logical structure of intensional contexts. This is the boundary case for the schema: the mediator is a context, not a form, and the stabilizing source is the formal-semantic structure of the embedding.

The list isn't closed. Other linguistic profiles may be stabilized by combinations of these or by sources the paper doesn't invoke (information-structural pressures, learnability constraints, normative/pedagogical regularization). The point is that "what stabilizes this profile?" is the right empirical question, and the answer is heterogeneous.

5.4 SPC and Khalidi as diagnostics

SPC and Khalidi are complementary diagnostics rather than nested ones. Slater asks whether the cluster is stable and projection-supporting and is neutral on *what* stabilizes it. Khalidi asks whether the stabilizing structure is itself causal-hierarchical, with named candidate core properties whose causal action generates derivative properties. A profile can satisfy SPC without satisfying Khalidi, and (in principle) the reverse. The projectibility profile is the unit of analysis; the two diagnostics ask different questions of it.

Slater (2015) gives the stability-plus-projection diagnostic. A profile satisfies SPC when its cluster of properties is stable enough to support reliable induction, regardless of *what* maintains the stability. Slater's framework has affirmative content beyond maintenance-neutrality: cluster stability has to be cliquish (members co-occur reliably under counterfactually relevant conditions), and sub-cluster-to-cluster probabilistic entailment is what supports reliable projection. Number-transparent quantificational nouns are the strongest candidate to satisfy SPC examined here: the cluster is stable, the sub-group covariance documented in §2.4 (determiner-pattern, partitive availability, override-direction co-vary by sub-group) is candidate evidence for the cliquishness SPC requires, and the maintenance is at least conventional/lexical for class identity (§5.3). The within-domain audit beyond override direction is empirically open; the cliquishness claim is licensed by the §2.4 sub-group covariance but not yet by an end-to-end corpus or experimental study.

TFRs are a second syntactic candidate. The agreement/syntactic-category split and the verb-class restriction form a projected cluster stabilized by constructional inheritance, though the present evidence is a working corpus check rather than a completed profile audit. Each form–meaning window (§3) is a candidate to satisfy SPC within its own field, pending the within-domain audits. Whether the family as a whole satisfies SPC at the cross-domain level is the open empirical question §5.2 flags.

Khalidi (2018) applies where the stabilizing structure is ordered and causal-hierarchical, with named core properties whose causal action generates derivative properties via identifiable pathways. QN class identity isn't such a case: it's conventionally stabilized without a separate ordered causal hierarchy, so SPC, not Khalidi, is the diagnostic that fits it. One level out, the count cluster the QN's *of*-complement feeds (*a lot of money* non-count, *a lot of tourists* count) has been argued to be causal-hierarchical, with conventional individuation as a candidate core property (Reynolds, 2026a). That analysis belongs to the countability paper and isn't needed here: the within-NP percolation the QN profile relies on can be observed and predicted under SPC alone. Khalidi names the vocabulary for asking whether a sub-profile has internal causal ordering; it isn't a result this paper delivers for the QN case.

For the form–meaning family of §3, Khalidi is acknowledged as a possible refinement *if* corpus predictors, rated transparency, priming magnitude, and interpretation accuracy align on shared items as joint reflexes of underlying representational structure. The four windows aren't a temporal chain (masked priming probes early lexical access before explicit interpretation, and ratings are a late meta-linguistic judgement) but candidate joint consequences of a single underlying organization. The within-items evidence doesn't yet exist; separate work sketches a within-items probe design. Until that probe is run, Khalidi remains a possible refinement for the form–meaning family rather than a delivered analysis.

Kindhood, on this picture, is the downstream verdict. A profile that scores well on the SPC diagnostic, and especially one that also scores well on the Khalidi diagnostic, has earned the kind label. A profile that scores modestly is a candidate. A profile that fails either diagnostic isn't a kind, even if it remains a useful descriptive category. The diagnostic work is *what stabilizes projection*; the kind label follows.

The apparatus matters in two places. SPC's maintenance-neutrality lets the paper put conventional, entrenched, architectural, and formal-semantic stabilizers on the same footing, where Boyd's homeostasis would force one mechanism (a poor fit for lexical convention). Khalidi's causal-hierarchy diagnostic gives the paper a vocabulary for asking whether sub-profiles have internal ordering: the form–meaning family's possible cross-modal cluster is the candidate case.

5.5 The resulting claim

Linguistic TRANSPARENCY isn't a kind, and the umbrella schema doesn't aspire to be one. Applying the label predicts nothing on its own: that something is *transparent* says the schema's slots are filled, not what the filling lets you infer, so what supports induction is the specific profile a transparency claim points to, not the word. The schema is a diagnostic tool for locating field-relative projectibility profiles: cases where an observable base licenses non-trivial projection to a target, stabilized by some identifiable source. Different transparency profiles are stabilized differently (lexical convention, constructional inheritance, usage entrenchment, processing architecture, formal-semantic environment), and each profile's projectibility is what makes the transparency attribution scientifically useful, not the profile's classification within any single ontology of kinds.

Three levels stay distinct, per §4:

– COMPARATIVE CONCEPT: mediated accessibility, the cross-linguistic methodological tool defined by the schema of §1.

– Descriptive category: language-particular grammatical content. English number-transparent quantificational nouns are a *CGEL* descriptive category; English NN compounds and *-ness* nominalizations are descriptive categories defined by language-particular criteria; Korean case-stacking and scrambling are Korean-particular descriptive categories; and so on.
– Projectibility profile: an observable base that licenses non-trivial projection to a target, stabilized by some identifiable source, realized within or across language-particular descriptive categories. This is the projectible profile of §5.1: a maintained-and-projectible cluster, contrasted with a *class* (mere distributional grouping). Profiles that score well on the SPC and (where appropriate) Khalidi diagnostics earn the local-kind label as a downstream verdict.

The contribution here is to apply the projectibility-first move to a *relational* concept. Transparency is the relation X-mediates-P-for-R, not an entity. Treating a relation this way needs something a property-kind analysis doesn't: a way to tell genuine mediation from cases where no property is being made accessible at all. The non-instance requirement of §2.3 supplies it, which is why the projectibility-first move transfers to a relational concept instead of dissolving into a list of co-occurrences. Asking *when does this attribution support induction, and what stabilizes the induction?* works for relations as it does for the umbrella concepts *language* and *grammatical category* addressed in related work. The paper asks that naturalization question of transparency too, and answers it locally, profile by profile, rather than at the umbrella level.

5.6 Connections and limits

Other multi-sense umbrella concepts (e.g. *language*, *grammatical category*) admit the same projectibility-first move: the umbrella isn't itself a kind but specific senses can be projectible profiles. Reynolds (2026b, ch. 7) develops the structural analogue for negative-polarity items at length.

The Korean (§2.6) extension is the most consequential open test. If mediated accessibility fails to yield productive cross-linguistic generalization when applied to Korean case-stacking and scrambling, the typological discipline of §4 admits the negative result without sacrificing the comparative-concept frame.

6 Conclusion

Linguistic transparency is a family of mediated accessibility relations, not a single syntactic property. The umbrella schema (X *is transparent with respect to* P *for* R *when* P*, which* X *might otherwise have concealed from* R*, remains accessible through* X *for the purposes of* R) is a comparative concept: an analyst's question, applicable to any language, asking which configurations let a property remain available to a grammatical or interpretive relation across a mediator.

Its distinctive contribution is a non-instance requirement: a configuration qualifies only when the mediator's own default behaviour could have concealed P from R. That filter separates genuine mediation from compositional summation and from everyday clarity, keeping the schema from relabeling familiar facts. It doesn't claim that any language has a descriptive category called transparency.

The main field-relative result is syntactic. Number-transparent quantificational nouns are the strongest candidate examined: a small lexical class whose membership predicts a cluster of correlated behaviours (agreement override, determiner patterns, partitive availability, fused-head behaviour, count/non-count percolation, restricted premodification), stabilized by conventional lexical inheritance. Transparent free relatives supply a second syntactic profile, with constructional sub-type and verb-class membership predicting agreement and syntactic-category transparency.

The form–meaning accessibility family (constructional compositionality, rated compound transparency, masked-priming morphological transparency) gives three adjacent within-domain profiles whose cross-domain integration on shared items remains an open research programme, developed in separate work that designs the within-items probe to settle it. Coordination is excluded as a non-instance; referential transparency is acknowledged as the schema's boundary case (§3.1).

The typological cases widen the schema beyond outward escape from an internal property. English QNs, TFRs, Spanish partitives, and Tsez long-distance agreement instantiate outward endpoint access; Aleut gives a qualified positive for null-source subpart access; Lardil and Kayardild show inward dependent access; and Chamorro wh-agreement shows path-distributed access. Algonquian/Kutenai obviation marks the negative boundary: when a feature is transformed or relayed onto a head, the predicate may track the head's derived feature rather than directly accessing the original source.

Those additions widen the topology without promoting every case to a worked projectibility profile. Aleut matters most for the original head-marking question, but Merchant's analysis keeps it distinct from overt intact-NP transparency. Lardil, Kayardild, and Chamorro pressure-test directionality and distribution of R while leaving QNs and TFRs as the paper's core syntactic profiles. Korean case-stacking and scrambling (§2.6) remain deferred for separate co-authored development. Acquisition trajectories are a plausible extension the paper doesn't ground.

A transparency attribution is scientifically useful when it identifies a field-relative projectibility profile: an observable base that licenses non-trivial projection to a target, stabilized by an identifiable source. Some profiles support strong projection; others weak; some none. Kindhood is the downstream verdict on profiles that earn it, not the master question of the analysis.

A Number-transparent NP empirical apparatus

This appendix collects the COCA pilot data behind the §2 claims about the number-transparent quantificational noun class. The body of the paper carries the qualitative claims; the audit trail lives here. Full machine-readable data: `paper/data/coca-pilot.md`; KWIC spot-checks: `paper/data/kwic-checks.md`.

A.1 Methodology

Queries were issued through the local Playwright wrapper at `tools/english-corpora/bin/ecorg.mjs` (May 2026). Each query is a surface string; counts are raw COCA frequencies, not filtered for head versus modifier use, idiom contamination, or sense disambiguation, except where noted. Two filtering issues recur:

- Modifier confound. Surface strings of the form X *of* N may include cases where N is a modifier inside a larger NP rather than the head of the *of*-complement (e.g., *lots of day-care* rather than *lots of day*). The countability paper (Reynolds, 2026a) handles the same coding issue for *three police* and *three cattle*. Where small counts fall in the predicted-counter direction, the modifier confound likely inflates them.
- Parse-shift. Surface strings containing a number-transparent QN plus a verb form may include cases where the QN sits inside a relative clause modifying a different head-noun, with the visible verb agreeing with that other head. *A lot of money are* is the textbook case: of 13 raw plural-agreement hits, 10 are parse-shifted (people/parents/movies/times *are*; *a lot of money* in a relative clause), 1 is irrealis *were* (*CGEL* §3.5.1), 1 *'s* hit is contracted *has* (singular, matches prediction), and 1 is a genuine plural override.

The wrapper's KWIC mode is unreliable for API-driven retrieval from this script as of May 2026 (Playwright internal error on every attempted call), but that doesn't mean KWIC is unavailable for human use. Counts in the tables below were generated via list-mode queries; KWIC checks were then used to filter selected small cells where raw surface counts were likely to overstate the target construction.

A.2 A.2 Partitive vs. pseudo-partitive *of*-complements

The contrast: partitive *of*-complements have definite complements (*of the money*, *of those books*); pseudo-partitive *of*-complements have bare or otherwise non-definite complements in the QN frame (*of money*, *of soldiers*, *of some paintings*).

A.2.1 Plural-only forms (*lots, bags, heaps, loads, oodles, stacks, piles*)

Pseudo-partitive (X of N): *lots of people* 3,878; *lots of money* 1,376; *lots of time* 703; *loads of money* 109; *piles of money* 91; *loads of people* 90; *loads of time* 34; *heaps of money* 15; *heaps of people* 7; plus single-digit hits for several other combinations. Total: 6,311. KWIC checks of the single-digit cells show the expected mixture: *lots of day* is modifier use (*day-to-day*, *day hikes*, *day trips*); *heaps of time* and *piles of time* are genuine pseudo-partitive mass-complement uses; *piles of people* is literal piling rather than the quantificational use.

Partitive (X of the N): *lots of the time* 7; *lots of the people* 5; *loads of the day* 1. Total: 13. *Piles* and *heaps* return 0 partitive hits with any complement tested.

Indefinite-NP complement (X of a N): *lots of a day* 0; *piles of a day* 0; *heaps of a day* 0; *loads of a day* 0; *lots/piles/heaps/loads of an afternoon* 0. Total: 0. Plural-only forms reject *of* [Det NP] whether the determiner is definite (*the*) or indefinite (*a/an*); their *of*-complements are bare/non-determined.

Partitive ratio: $13/(6{,}311 + 13) \approx 0.2\%$.

A.2.2 *Plenty*

Pseudo-partitive: *plenty of time* 4,017; *plenty of people* 1,563; *plenty of money* 668; *plenty of day* 4 (likely modifier). Total: 6,252.

Partitive: *plenty of the time* 1; *plenty of the people* 1. Total: 2.

Partitive ratio: $2/6{,}254 \approx 0.03\%$.

A.2.3 *Rest, remainder*

Pseudo-partitive (the X of N): *the rest of time* 75; *the rest of people* 12; *the rest of day* 7; *the remainder of time* 3; *the remainder of day* 2; *the rest of money* 1. Total: 100 raw. KWIC checks confirm that this raw total is conservative. *The rest of time* is mostly fixed or duration-level temporal use (*for the rest of time*, *spend the rest of time*), not the partitive-count contrast targeted here; *the rest of people* contains possessives and non-standard "other people" uses; *the rest of day* is temporal ellipsis or *day* as a modifier; and the *remainder* hits are technical or duration expressions.

Partitive (the X of the N): *the rest of the day* 1,932; *the rest of the time* 830; *the rest of the people* 272; *the rest of the money* 222; *the remainder of the day* 75; *the remainder of the time* 21; *the remainder of the money* 13. Total: 3,365.

Indefinite-NP complement: *the remainder of a day* 1.

Partitive (definite NP) ratio: $3{,}365/(3{,}365 + 100 + 1) \approx 97.1\%$ raw, higher after KWIC filtering of the non-target temporal, possessive, and generic uses in the raw pseudo-partitive cells.

A.3 Override-direction agreement

Test of the "near-obligatory in central cases" claim. Subject NP plus finite verb agreement, by complement number/count.

Subject	Predicted	Plural agr.	Singular agr.	% predicted
a lot of people	plural	4,195 (ARE 3,406; WERE 789)	85 (IS 72; WAS 13)	98.0%
a lot of money	singular	1 (ARE, filtered)	90 (IS 61; WAS 29)	98.9%
a number of people	plural	100 (ARE 52; WERE 48)	0 (4 false positives)	100%
lots of people	plural	348 (ARE 288; WERE 60)	0 (6 false positives/non-standard)	100%
lots of money	singular	0 (1 false positive)	24 (IS 19; WAS 5)	100%
plenty of people	plural	79 (ARE 65; WERE 14)	0	100%
plenty of money	singular	0	6 (IS 4; WAS 2)	100%
the rest of the people	plural	19 (ARE 15; WERE 4)	0	100%

Subject	Predicted	Plural agr.	Singular agr.	% predicted
the rest of the money	singular	0	19 (IS 10; WAS 9)	100%

Override is in the predicted direction at 98–100% across all tested QN+complement combinations after filtering. Of these, the discriminating cells, where the head's number and the complement's diverge (*a lot of people*, *a number of people*, *lots of money*, *the rest of the people*), are what evidence the override; the matched cells (*a lot of money*, *lots of people*, *the rest of the money*) confirm the predicted direction but are equally consistent with head-driven agreement, so they don't discriminate. The filtering below was applied to counter-direction cells; predicted-direction cells were not hand-scrubbed to the same depth, so these figures are exploratory corroboration of the categorical override, not estimated rates. *A lot of money* required filtering: of 13 raw plural-agreement hits, 10 were parse-shifted (the surface string *a lot of money are* appearing inside relative clauses modifying a plural-noun head, e.g., *people who earn a lot of money are successful*; *movies that make a lot of money are the biggest help*; *times when we... make a lot of money are gone*), 1 was the irrealis *were* (*if a lot of money were at stake*), and 1 *'s* hit was singular *has* (matches prediction); only 1 hit was a genuine plural override (*a lot of money are being raised*). Filtered count: 1 plural / 90 singular = 98.9% singular.

The other checked counter-direction cells are artefacts. *A number of people is/was* is headed by *condition*, *suggestion*, *ability*, or *counsel*, not by *number*; *lots of people is/was* is headed by expressions such as *murder*, *hauling*, or *disappearance*, with one non-standard existential hit; and *lots of money are* occurs inside a PP modifying plural *men*. The large *a lot of people is/was* cell remains unfiltered, so the 98.0% row is conservative.

A.4 A.4 *Bunch*: animate vs. inanimate aggregates

CGEL claims that *bunch* tilts collective for inanimate aggregates and transparent for groups of people. The COCA pilot supports the animate half but not the inanimate half.

Subject	Total tokens	*was*	*were*
Animate aggregates			
a bunch of people	1,421	0	28
a bunch of kids	448	0	12
a bunch of hooligans	12	0	0
Inanimate aggregates			
a bunch of flowers	68	0	0
a bunch of cars	–	0	1
a bunch of papers	–	0	0
a bunch of leaves	–	0	0
a bunch of things	–	0	1

Animates: 40 plural / 0 singular = 100% plural. KWIC checks found no parse-shift confound in these animate-plural hits. The *CGEL* "transparent for groups of people" claim is supported.

Inanimates: 2 plural / 0 singular. The *CGEL* "tilts collective for inanimate aggregates" claim isn't supported in COCA. Inanimate-bunch is mostly absent from subject position (typical use is as object: *she gave him a bunch of flowers*); the two inanimate-bunch hits in subject position both took plural rather than the *CGEL*-predicted singular. *A bunch of flowers was presented to the teacher* (the *CGEL* example) returns 0 hits in COCA.

A.5 Negative control: referential *the number of*

If the override tracked the quantificational construction rather than the noun *number* or N-of-N configurations in general, then referential *the number of*, where *number* is the head rather than a quantifier, should agree head-driven (singular) even with a plural complement. A COCA check (June 2026) bears this out on the same lexeme. *The number of people* occurs with *is*/*are* at 3/1 and *the number of cases* at 20/1: head-driven singular, with plural only at attraction level. *A number of people* occurs at 0/52 and *a number of cases* at 0/3: categorical plural override. The counts are small, since the bare subject-verb adjacency is rare, but the direction is unambiguous and the contrast sits on one noun: *number* overrides under *a* and not under *the*. The override is therefore a property of the quantificational construction, not of the lexeme or of N-of-N structure as such.

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